

Estimation of Wheel-Rail Contact Patch Geometry Under Curving Conditions

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ABSTRACT

This study evaluates the two-point contact geometry that occurs in a curve using the Virginia Tech Federal Railroad Administration (VT-FRA) roller rig, which employs a one-fourth scaled AAR-2A wheel and 136RE rail. The contact patch geometry and pressure distribution across it are measured using pressure-sensitive films. This study provides a fresh perspective on the understanding of the complex mechanics and dynamics of the two-point contact at the wheel-rail interface (WRI). Specifically, it evaluates how the distribution of tread and flange contact changes for different angles of attack (AoA), rail cant, and lateral position of the wheel relative to the rail beyond the simulation studies that have been offered in the past. Furthermore, the distribution of lateral forces at the flange and vertical forces at the tread are measured to assess the change in Lateral-to-Vertical (L/V) forces that are often used as an indicator for wheel climb derailment in curves.

An initial set of simulations is done using CONTACT, a WRI simulation software, which shows excellent agreement within a 25% difference. The experiment parameters are carefully recorded during the tests to ensure accurate implementation of the simulation model under the same parameters including the actual wheel and rail profiles. Additionally, the model is fine-tuned by adjusting the mechanical properties of the mylar-based film used in the actual measurements to obtain an acceptable agreement between the tests and simulations. This will help validate the numerical algorithm in which the software is built.

The results highlight the location and centroid of contact pressure are highly affected by the relative lateral position and angle of attack between the wheel and rail. With a large lateral force, the contact area (single point) is maximum for a moderate Angle of Attack, although observed in lower lateral forces with a lesser peak. At a large Angle of Attack where two-point contact is followed, there is an equal area distribution between the contact points. Additionally, there is almost a constant area distribution at the tread at different lateral forces. It shows the intricacies of wheel-rail interaction and the factors that may

cause derailment. This study helps validate the contact model used and proves to be an important step toward the dynamic research of wheel climb derailments.

Introduction

The interaction between the wheel and the rail is a critical aspect of railroad engineering (1–4). Although several researchers have consistently progressed in understanding and evaluating wheel-rail interface (WRI) in a tangent track, there has been limited progress in a curving condition. Currently, the railroad industry relies on the extensive experience of railway operators and engineers and numerous analytical and numerical contact models to simulate a wide range of field conditions, ultimately enhancing the safety and reliability of railcars. The inaccessible nature of the interface has made it difficult for researchers to understand and calibrate these models.

The published mathematical and analytical models (5–7) provide predictions on the contact geometry and pressure distributions under normal contact conditions, some of which follow the Hertzian contact model. The railroad industry widely uses the Hertzian contact theory (8,9) within its limitations. The Hertzian contact theory assumes a fixed single-point contact, the contact radii are always larger than the ellipse semi-axes, and the curvatures are constant as the contact point. Thus, it does not truly represent the actual shape with changes in the wheel profile and two-point flange contact. Several non-hertzian contact models have also been theorized, such as the contact algorithm by Polach (10), Piotrowski (11,12), and Kalker's non-Hertzian models (13). The 3D rolling theory developed by Kalker (14), used in CONTACT software developed by Vtech-CMCC (15), provides the best results and is considered a benchmark by other researchers and authors to solve complex wheel-rail contact problems. An overview of various contact models is available in studies such as (16,17). These contact models are developed to dynamically simulate normal or tangential wheel-rail contact, i.e., to estimate creep forces, longitudinal and lateral creepages, etc. Continuous progress is being made to improve these models and theories. These contact models are developed to dynamically simulate normal or tangential wheel-rail contact, i.e., to estimate creep forces, longitudinal and lateral creepages, etc.

There is a need to assess the normal and tangential contact experimentally. For decades, several notable researchers have used multiple means to capture the contact patch between the wheel and the rail. New novel ultrasound devices (18–23), carbon paper (24), and pressure-sensitive films (25,26) are some of the experimental approaches adopted to capture the contact patch measurements statically. The ultrasound devices have provided results that do capture the contact patch, but the test rigs used do not replicate the field conditions. This involves much effort in a custom structure for adapting the measurement units to each test rig, making it not readily feasible. On the other hand, carbon paper does not provide any understanding of the pressure distribution at the contact.

This paper employs three distinct pressure-sensitive films, like those previously used (27), to estimate the geometry of the contact patch and pressure distribution in a curve while in a static state. This approach provides a continuation of the steps taken to explore the contact patch pressure distribution but for conformal contact at the throat and two-point contact as a result of large AoA and lateral forces. The Virginia Tech-Federal Railroad Administration (VT-FRA) Roller Rig accurately simulates curving conditions to conduct these static experiments. The Angle of Attack (AoA), lateral positions, and rail cant are altered to emulate curving conditions while keeping the applied wheel load constant. The main purpose of this paper is to provide the results of static evaluations that can experimentally verify the existing contact models and software such as CONTACT (15,28,29).

Test Setup

The Virginia Tech—Federal Railroad Administration Roller Rig shown in (**Figure 1**) is employed for static assessment of the contact patch geometry under curving conditions. The 1/4th-scale roller rig can precisely mimic the contact statics and dynamics of the wheel-rail interface (WRI). A high precision laser scanner is used to measure the surface profile and third-body layer accumulation at the contact. Furthermore, the rig includes six linear actuators that can accurately control the vertical and lateral positions, as well as the angle of attack (AoA) and rail cant angle. It also has two servo motors that precisely control the speed of the wheel and roller individually, hence enabling testing in various %creepage conditions.

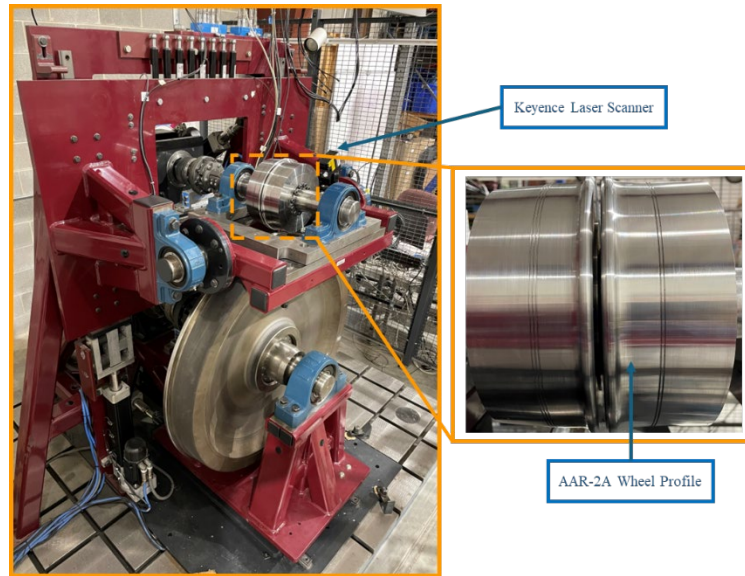


Figure 1. Virginia Tech – Federal Railroad Administration (VT-FRA) Roller Rig

To replicate curving conditions, the AoA can be set from -6° to $+6^{\circ}$ and Rail Cant from -3° to $+3^{\circ}$. The settings allow testing in curving conditions commonly observed in the field.

An AAR-2A wheel profile with a nominal diameter of 10 inches is used for this research. The AAR-2A wheel profile is designed to undergo lesser flange wear by eliminating or reducing the intense two-point contact observed with other wheel profiles. The wheel profile has a flange angle of 75° and a 1:20 (3°) taper. The roller profile on the roller rig is a standard US-136 rail profile with a diameter approximately five times larger than the wheel. This minimizes the contact patch distortion to less than 10%. Additionally, this helps extend the rig's results to the field by a length scale of four (30).

Six different Angles of attack and two distinct cant angles commonly observed in the field were selected to emulate the field conditions. Derailment ratios, i.e., lateral to vertical force ratios, were used to precisely control the vertical and lateral forces at contact to establish the force conditions. The linear actuators on the roller rig can control this with utmost precision. 0.25 to 1.00 with an increment of 0.25 was considered a suitable L/V ratio for the static tests.

Forty-eight sets of experiments were conducted at 5.0 kN vertical load. The vertical force applied can also be scaled by a factor of 16 to extend it to the field. A detailed description of the capabilities and scaling factors can be found in (30). (Table 1) comprises the summary of experiments designed for these static test conditions. Vtech-CMCC's CONTACT will also simulate the static test measurements to understand the software's capabilities.

Table 1: Test Conditions

Test Parameter	Value
Wheel Profile	AAR - 2A
Roller Profile	US-136 rail
The angle of Attack (Yaw) ($^{\circ}$)	0 to 5 (increment of 1)
Cant Angle (Roll) ($^{\circ}$)	1.5 and 3
Rig Wheel Load (kN)	5.00
Full-Scale Wheel Load (kN)	80.00
Lateral and Vertical Force Ratio	0.25 to 1.00 (increment of 0.25)
Measurement Mode	Static

Pressure-Sensitive Films

Fujifilm prescales® pressure-sensitive film, a mylar-based film, is used for contact patch estimation. The film contains a layer of tiny microcapsules. These small particles rupture under high contact forces, producing an instantaneous and permanent high-resolution image. The working method can be seen in **(Figure 2)**. This image accurately shows the topographical image of pressure distribution at the contact area with different color intensities (31).

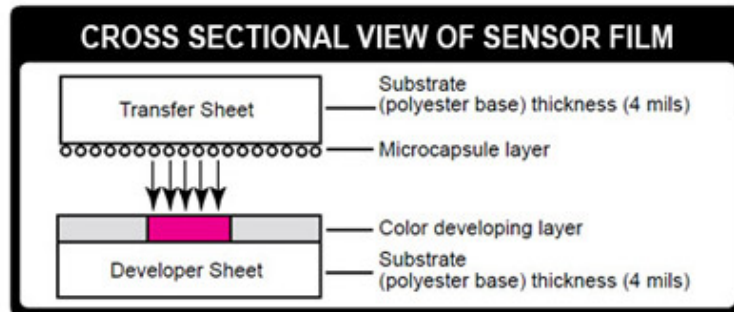


Figure 2: Fujifilm Prescale Sheet Working Method (31)

The contact patch geometry is estimated using various grades of pressure-sensitive films, like what was previously done on the VT-FRA Roller Rig (27). **(Table 2)** provides the three types of films used to determine the contact geometry and pressure distribution.

Table 2: Type, Pressure Range, and Thickness of Pressure-sensitive Films

Film Type	Pressure Range	Thickness
Super Low	70 - 350 psi (0.5 - 2.4 MPa)	Two Sheets: 90 μ m each
High	7100 - 18500 psi (49.0 - 127.6 MPa)	Mono Sheet: 100 μ m
Super High	18500 - 43200 psi (127.6 - 297.9 MPa)	Mono Sheet: 100 μ m

The pressure range associated with each type determines the span within which the color distribution on the film is detectable. When the pressure applied is below the lower limit of this range, the film remains colorless. Conversely, pressure exceeding the upper limit results in the most intense color display, with the film showing the same intensity for pressure beyond the upper limit.

All experiments are performed on the wheel and roller profiles at the same location, allowing for comparable contact patch shapes in various test cases. It is essential to mention that the local plastic deformation is ignored in these considerations. Each type of film is placed on the wheel surface and taped meticulously. Once this is done, the wheel and the roller are brought into contact to apply the required wheel load. **(Figure 3)** shows the before-and-after images of the contact patch seen on the Roller Rig. The left contact point, shown in **(Figure 3c)** represents the flange contact point. Similarly, the right contact point represents the tread location.

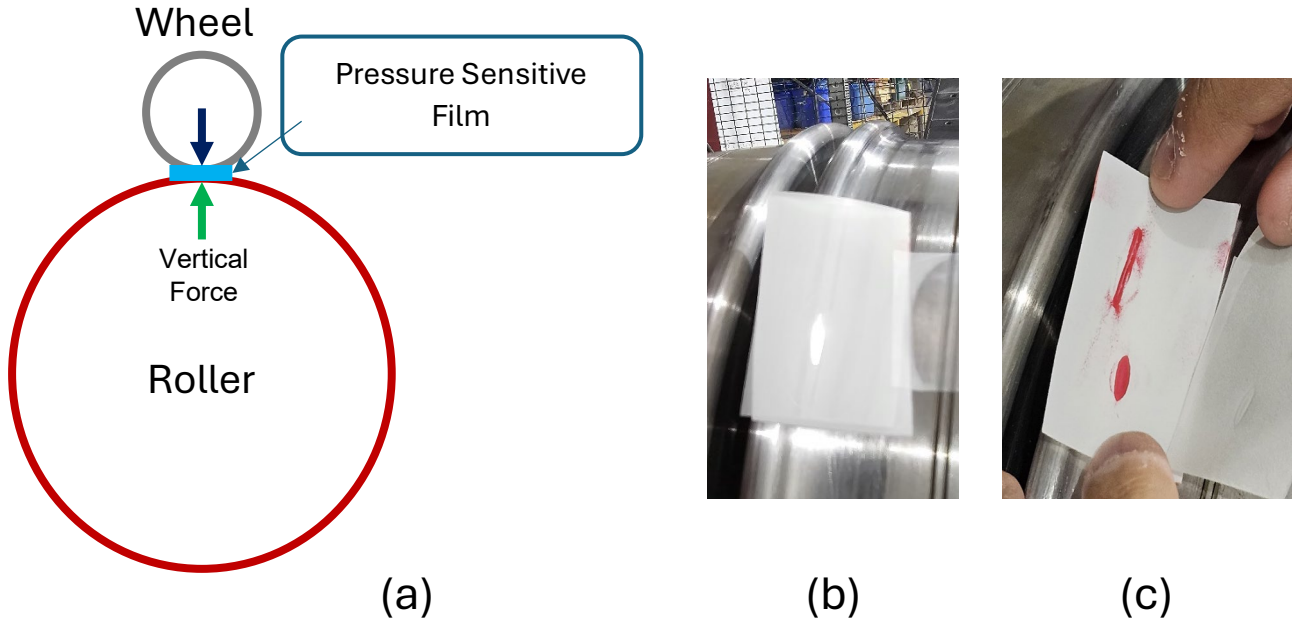


Figure 3: (a) Front View of Contact Patch estimation method on the VT-FRA Roller Rig, (b) Before Experiment, (c) After Experiment

Results and Discussions

The tapered section of the AAR-2A wheel profile has a 1:20 taper, corresponding to an approximate roll angle of 3° . Half the experiments are performed with a rail cant of 3° towards the vertical axis, so the wheel load applied would equal the normal force at contact. The other half of the tests are performed with a rail cant of 1.5° , corresponding to a contact angle of 1.5° .

The results of this study are divided into three sections: Low (0° & 1°), Medium (2° & 3°), and High (4° & 5°) AoA. To better clarify the contact patch observed, the pressure-sensitive films are scanned and colored. Green represents the super-low film range, yellow represents the high film range, and red represents the super-high film range.

(Figure 4) shows the results from the tests with low AoA where predominately single-point contact is observed. Each column represents the measurements under the same L/V ratio. Measurements in the same row represent the specific rail cant angle applied. The green coloration provides an approximate total contact area between the wheel and the rail. This is because super-low film can capture pressures as low as 0.5 MPa. On the other hand, super-high films do not capture pressures below 127.6 MPa; hence, a smaller area at contact is observed in red.

An AoA of 0° represents a tangent track; as expected, the contact patch is estimated to be elliptical, almost following the Hertzian contact model. As the roller moves laterally towards the flange, the L/V ratio increases, and the area of the contact patch also increases. At higher lateral positions, i.e., lateral forces, the contact is at the throat of the wheel profile, which can be observed by an increase in the area and the aspect ratio of the contact patch. A similar trend is observed even at rail cant 1.5° . The only observable change is the larger contact area. This is expected because of a large contact angle.

At an AoA of 1° , the normal contact observed at 3° rail cant show an increase in the aspect ratio. Noticeably, the contact patch area grows from the tread to the wheel's flange, thereby increasing the area of contact. At high lateral forces, the large contact pressure is distributed more at the throat of the wheel profile. Similarities are observed for rail cant 1.5° . It is important to note that no two-point contact is detected.

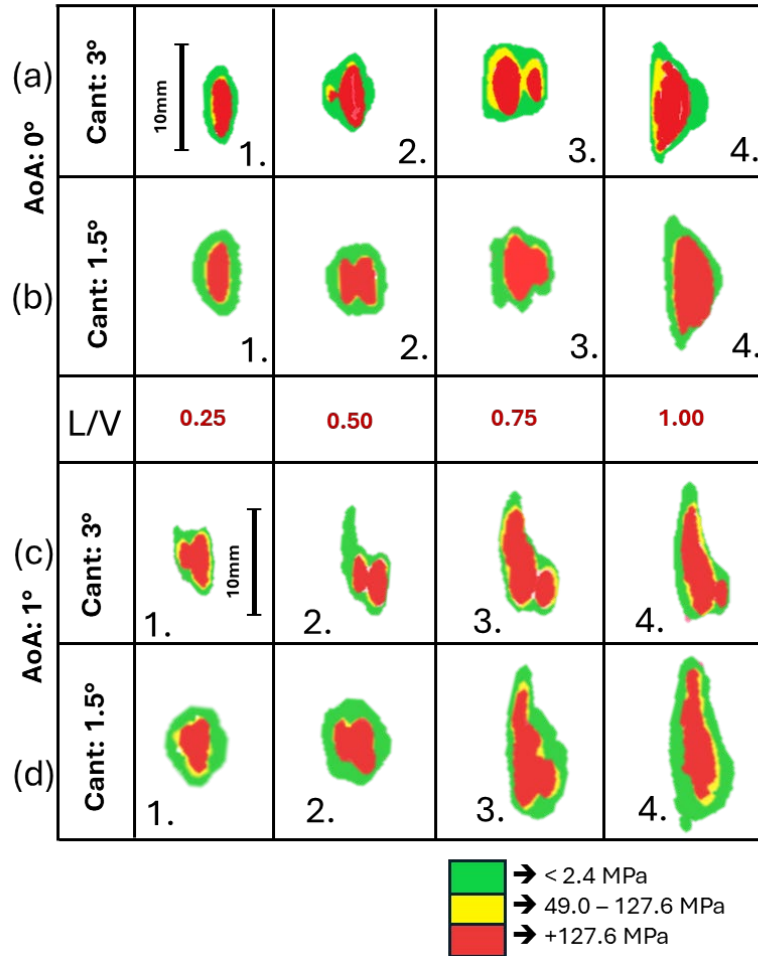


Figure 4: Contact patch measurements for Low AoA using pressure-sensitive films.

(Figure 5) presents measurements for the tests done with medium AoA. (Figure 5 e,1 and e,2) show that single-point contact is observed for lower lateral forces. As the lateral position increases, the contact patch grows towards the flange. The 'f' shown in the figure represents the flange contact point location. Similarly, 't' represents the tread location. The aspect ratio also increases. At larger contact angles, two-point contact is perceived with lower lateral forces. Large contact pressures are not observed on the flange contact point, which can be seen by the lack of red (Figure 5 f,1 and f,2).

At AoA of 3°, two distinct points of contact are noted. Similar observations about the aspect ratio are made as it increases with the increase in the L/V ratio. Visually, the tread contact patch maintains its geometry. With 1.5° cant, a similar trend is observed, but with a larger contact area. It is interesting to notice that the distance between the two contact points is approximately 10mm. This observation can be a tool for comparison in simulating these static tests with contact models such as CONTACT.

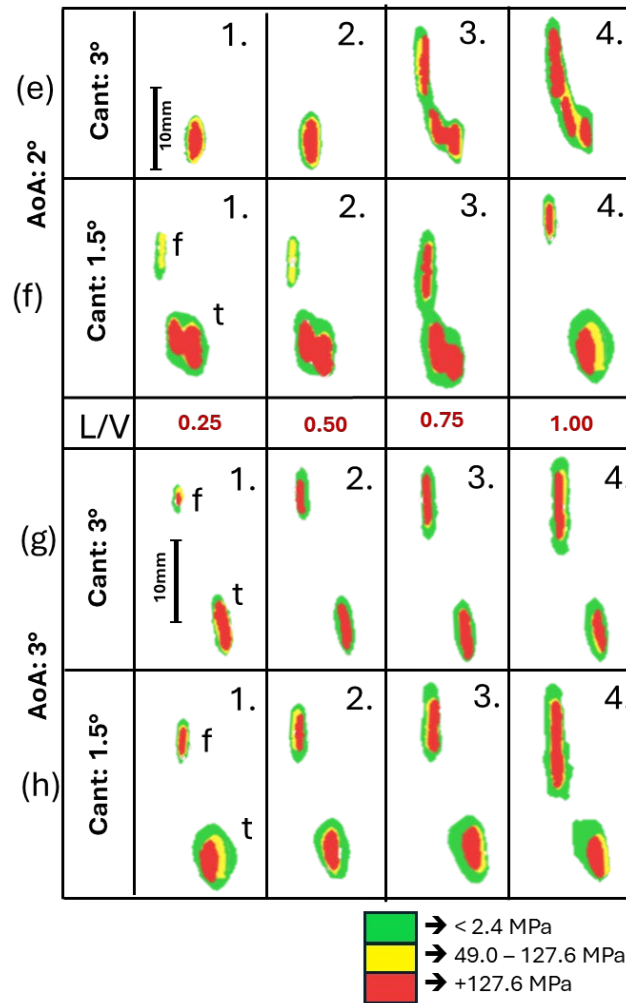


Figure 5: Contact patch measurements for Medium AoA using pressure-sensitive films.

(Figure 6) shows the results for High AoA. For this Angle of Attack, the two-point contact observed is very distinct. The 'f' and 't' illustrate the flange and tread contact locations. For lower lateral forces, the flange contact point is smaller than the tread contact. As the lateral position changes to increase the lateral force applied, the aspect ratio increases. The tread contact point area remains approximately the same across the L/V ratio. It is interesting to observe the grooved edges in most of the measurements and the additional third point that can be seen in (Figure 6 k,1). These measured observations are mainly due to the finishing of the wheel surface profile and past experiments performed toward the outer side of the tread region.

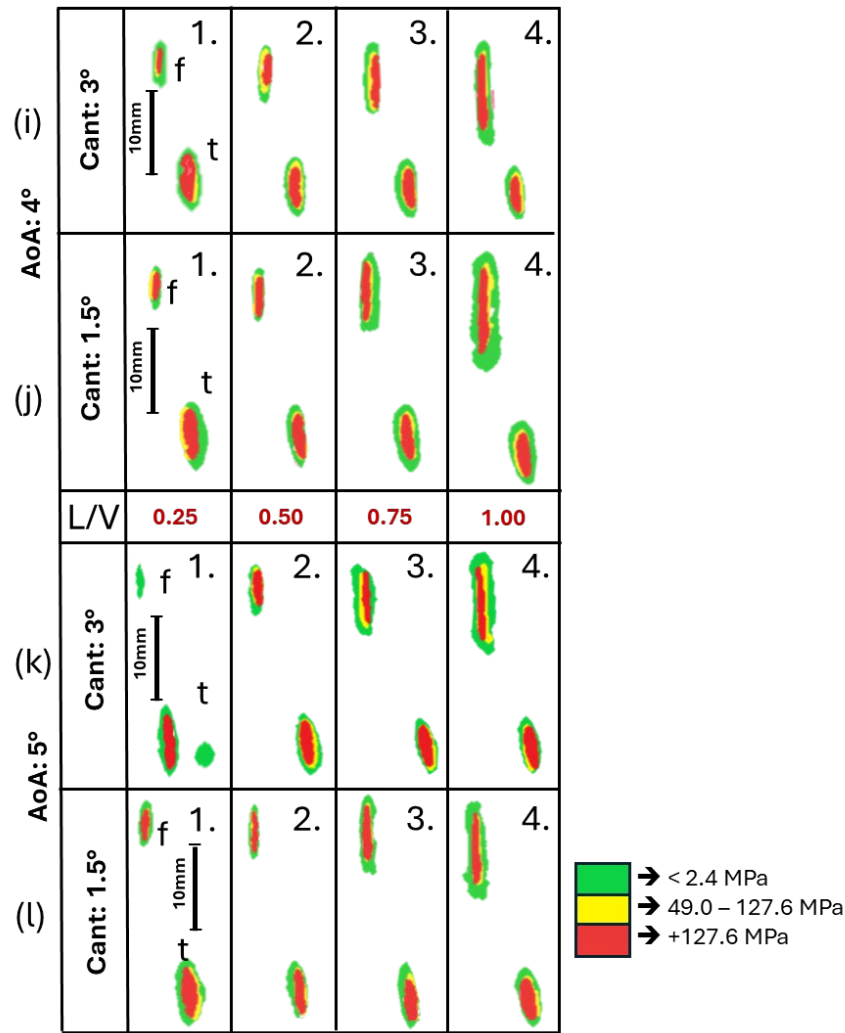


Figure 6: Contact patch measurements for High AoA using pressure-sensitive films.

A summary of the estimated area using pressure-sensitive film is shown in **Figure 7**. The results for all AoA and rail cant angles are summarized in (**Figure 7 a**) at 0.25 L/V ratio, (**Figure 7 b**) for 0.50, (**Figure 7 c**) for 0.75, and (**Figure 7 d**) for 1.00. However, as the AoA increases, there is a noticeable transition from a single-point contact to a two-point contact scenario. This transition is evident through a decrease in the contact area at the tread point and a simultaneous increase in the contact area at the flange point. This trend is consistent across varying L/V ratios, indicating a robust pattern influenced by the angle of attack.

Furthermore, the shift in contact points aligns with expected theories, where higher AoA induces a shift in force distribution, leading to increased contact at the flange. This observation underscores the significance of AoA in shaping the contact geometry between the wheel and rail, with implications for stability, wear patterns, and overall safety.

By analyzing these trends across different L/V ratios and AoA values, a comprehensive understanding of the contact patch dynamics under varying curving conditions can be gained.

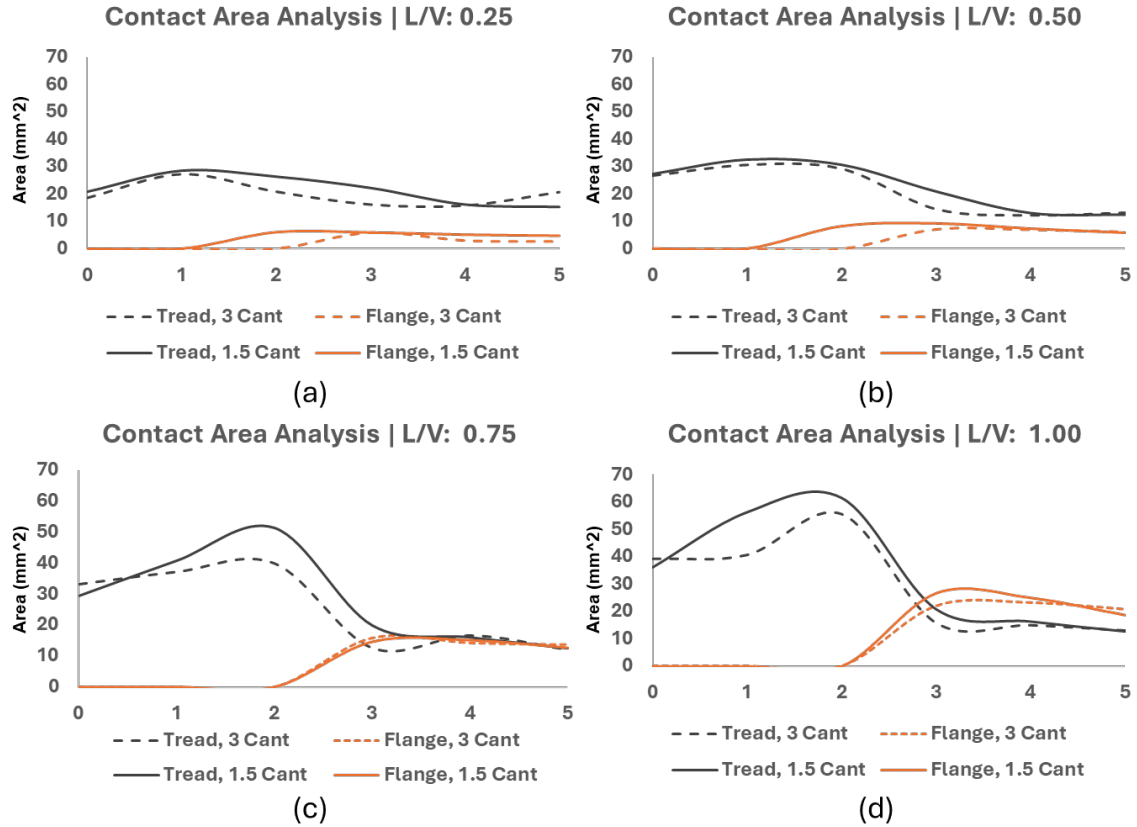


Figure 7: Contact Area Analysis for (a) L/V: 0.25; (b) L/V: 0.50; (c) L/V: 0.75; (d) L/V: 1.00

(Figure 8) shows the area analysis and the general trend observed for the tread and flange contact locations for higher AoA as the L/V ratio increases. The dotted black and orange lines represent the trend observed for the tread and flange contact points for the applied rail cant angle of 3°. Similarly, the solid lines of black and orange coloration represent a cant angle of 1.5°.

These area measurements provide a concrete quantitative understanding of what can occur in a curve. The area is compared to the L/V ratio, which can be analyzed as the roller's lateral position increases closer to the wheel's flange. Notice that the dotted and solid orange lines show an almost linear increase in the area. This increase also corresponds to the growth of the aspect ratio of the tread contact point.

There is almost no change in the tread contact point. The slight differences that can be seen may be due to the elastic effect of the film, which is seen to have little to no impact on the overall results. This was previously observed in (27) and showed the method of using pressure-sensitive film to be highly repeatable. This can be understood as an increase in the pressure distribution on the flange contact point than the tread contact point. The overall area of the tread contact point does not affect the increase or decrease in the case of large AoA. This is considered an important finding for the future design of experiments for the contact dynamics-related aspect of testing.

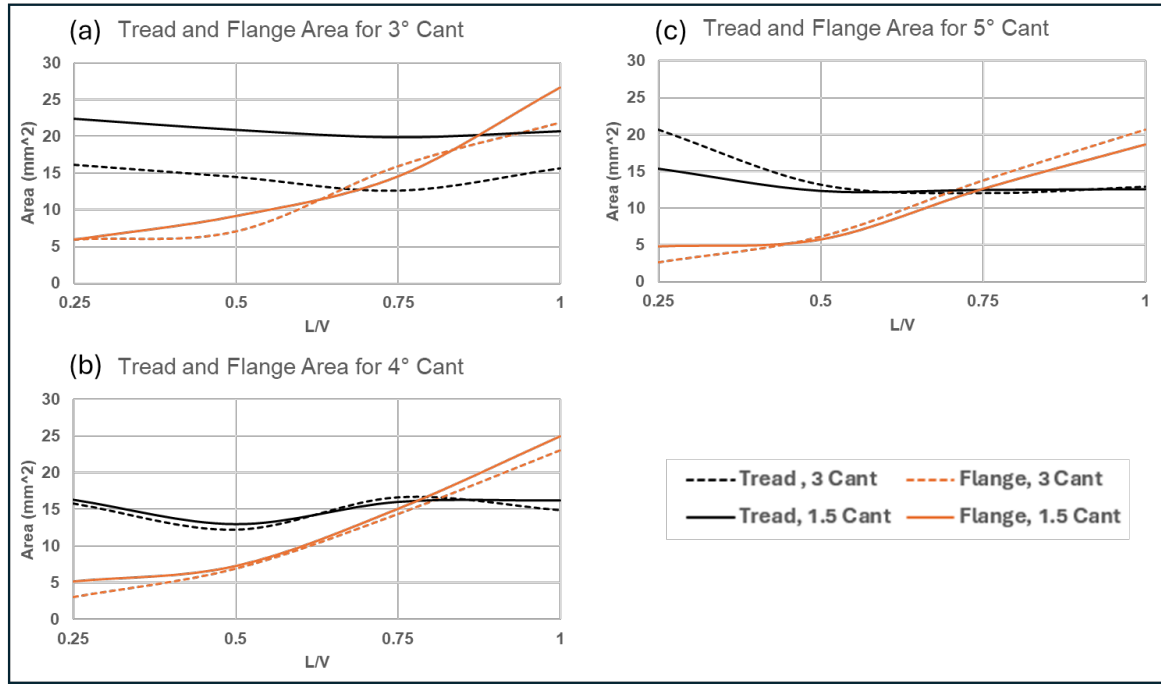


Figure 8: Area Analysis for flange and tread contact point at higher AoA

Contact Simulations

Vtech-CMCC's CONTACT software is used as a medium to replicate the static measurements taken in the VT-FRA Roller Rig. The input parameters used to run the simulations are the AAR-2A wheel profile and the roller profile scanned using the Keyence laser scanner shown in (**Figure 1**). This helps provide much more accurate results with the actual profile. Additionally, the lateral position (Y), wheel load (F_z), cant angle (Roll), and angle of attack (Yaw) are established in each simulation. The lateral displacement (Y) is controlled to adjust the lateral forces (F_y) such that the L/V ratio is maintained.

To incorporate the effect of film in contact, a flexibility (FLX) parameter is added in the script with the material and mechanical properties of the PET-based film.

The flexibility parameter provided for CONTACT is governed by (1)

$$Lz = h^{(3)} \frac{(1+\nu^{(3)})(1-2\nu^{(3)})}{(1-\nu^{(3)})E^3} \quad (1)$$

The paper highlights some preliminary findings that shed light on the abilities of the CONTACT software. The tests involved an AoA of 5° and a cant angle of 3° and are presented in Figure 9. The results show that using this simulation method for future testing and validation is promising, as the initial observations indicate a positive trend in the point of contact, which includes the jagged edges surrounding it, the third point of contact, and the overall geometry of the contact points. We do observe some differences in the area from the simulations. This may be due to the FLX parameter (1) used. Hence, this calls for tuning the model to get as few errors as possible.

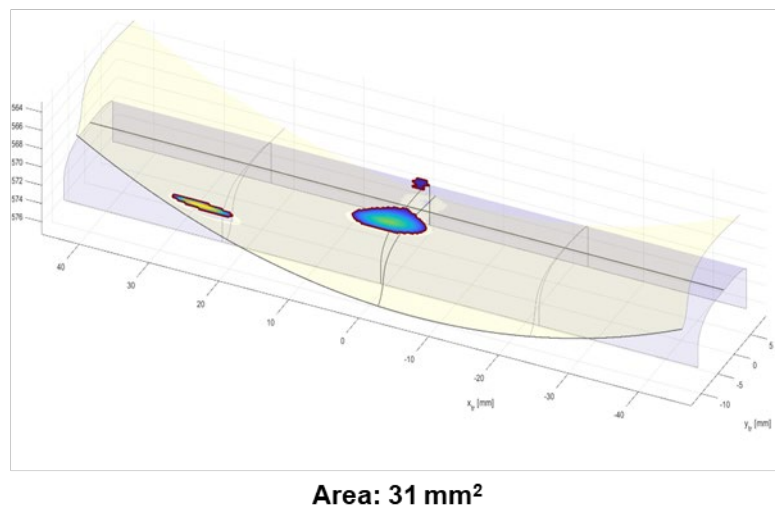
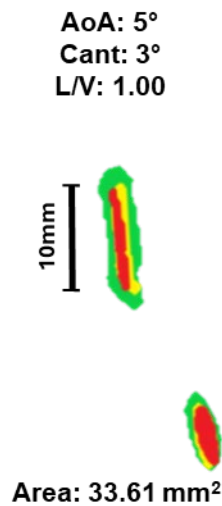
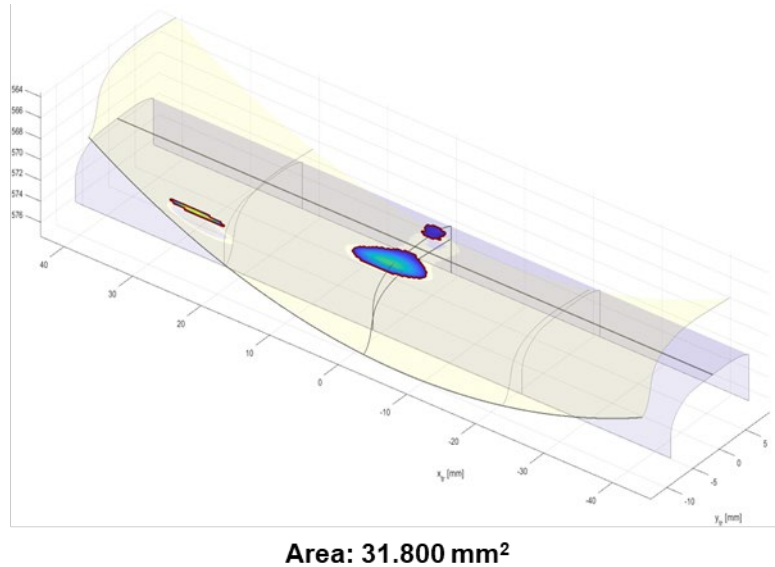
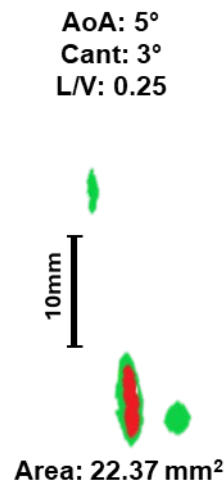


Figure 9: Initial CONTACT simulations to replicate static measurements.

CONCLUSIONS

This study evaluates the change in the wheel-rail interface (WRI) due to two-point contact for various angles of attack (AoA) and cant angles. Three pressure-sensitive films were used to estimate the two-point contact geometry and pressure distribution between the wheel and the rail. The resulting static measurements were evaluated for various AoA, lateral position, and cant angle on the Virginia Tech-Federal Railroad Administration (VT-FRA) Roller Rig to emulate curving conditions. For the large wheel loads applied during the tests, the measurement method proved to be effective in measuring contact pressure variations with great resolution.

The contact pressure at lower AoA appeared to be concentrated toward or at the wheel tread, as only a single point of contact was observed. When the cant angle was increased, the flange contact occurred, and the centroid of the contact moved toward the flange. With the increase in the lateral forces due to flange contact, the results show that the tread forces reduce and the flange forces increase, which is in line with the shift in pressure centroid toward the flange. The increase in flange forces is important not only because it increases flange wear but also because it poses a higher risk of wheel-climb derailment. The findings can also help with choosing the rail cant angle in curves so that the flange forces are reduced.

The results of the study also shed further light on the wheel-rail interface (WRI), the intricate relationship between contact geometry, lateral displacement, rail cant, and angle of attack on a curved track. The

test data can be used for tuning contact models in software such as CONTACT better to reflect the physical interaction between the wheel and rail. The tuning of the model will help in performing steel-to-steel simulations, which can be directly correlated to the field.

These findings significantly contribute to our understanding of the wheel-rail interface dynamics and their potential effect on derailments, thereby paving the way for more effective strategies to enhance train safety. The outcome of this study will also be used to extend the tests to dynamic conditions that will allow us to evaluate the dynamics of the wheel climb derailment better.

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References

1. Lewis R (Roger), Olofsson U (Ulf). Wheel-rail interface handbook. CRC Press; 2009. 842 p.
2. Lewis R, Olofsson U. Basic tribology of the wheel-rail contact. In: Wheel-Rail Interface Handbook. Elsevier Ltd; 2009. p. 34–57. doi: 10.1533/9781845696788.1.34
3. Hosseini S, Ahangarnejad AH, Radmehr A, Tajaddini A, Ahmadian M. A Statistical Approach to Evaluating Wheel-Rail Contact Dynamics. In: Joint Rail Conference. 2021.
4. Iwnicki S. Handbook of Railway Vehicle Dynamics. Iwnicki S, Spiryagin M, Cole C, McSweeney T, editors. CRC Press; 2019. doi: 10.1201/9780429469398
5. Kalker JJ. A Fast Algorithm for the Simplified Theory of Rolling Contact. Vehicle System Dynamics. 1982 Feb 1;11(1):1–13. doi: 10.1080/00423118208968684
6. Vollebregt EAH, Wilders P. FASTSIM2: A second-order accurate frictional rolling contact algorithm. Comput Mech. 2011;47(1):105–16. doi: 10.1007/s00466-010-0536-7
7. Sugiyama H, Suda Y. On the contact search algorithms for wheel/rail contact problems. J Comput Nonlinear Dyn. 2009 Oct;4(4):1–7. doi: 10.1115/1.3187211
8. H. R. H. Über Die Berührung Fester Elastischer Körper Und Über Die Härte. Verhandlung des Vereins zur Beförderung des Gewerbefleißes. 1882;449.
9. Knothe K. History of wheel/rail contact mechanics: From Redtenbacher to Kalker. Vehicle System Dynamics. 2008 Jan;46(1–2):9–26. doi: 10.1080/00423110701586469
10. Polach O. Fast wheel-rail forces calculation computer code. Vehicle System Dynamics. 2000;33(SUPPL.):728–39. doi: 10.1080/00423114.1999.12063125
11. Piotrowski J, Kik W. A simplified model of wheel/rail contact mechanics for non-Hertzian problems and its application in rail vehicle dynamic simulations. Vehicle System Dynamics. 2008 Jan;46(1–2):27–48. doi: 10.1080/00423110701586444
12. Piotrowski J, Chollet H. Wheel-rail contact models for vehicle system dynamics including multi-point contact. Vehicle System Dynamics. 2005;43(6–7):455–83. doi: 10.1080/00423110500141144
13. Kalker JJ. A simplified theory for non-hertzian contact. Vehicle System Dynamics. 1983;12(1–3):43–5. doi: 10.1080/00423118308968716
14. Kalkers JJ. The Computation Of Three-Dimensional Rolling Contact With Dry Friction. Journal For Numerical Methods In Engineering. 1979;14:1293–307.
15. Vollebregt EAH. User guide for CONTACT, Rolling and sliding contact with friction. 2023.
16. Meymand SZ, Keylin A, Ahmadian M. A survey of wheel-rail contact models for rail vehicles. Vol. 54, Vehicle System Dynamics. Taylor and Francis Ltd.; 2016. p. 386–428. doi: 10.1080/00423114.2015.1137956

17. Kalker JJ. Survey of Wheel—Rail Rolling Contact Theory. *Vehicle System Dynamics*. 1979 Sep 1;8(4):317–58. doi: 10.1080/00423117908968610
18. Marshall MB, Lewis R, Dwyer-Joyce RS, Olofsson O, Björklund S. Measuring Wheel/Rail Contact Stresses using Ultrasound. Available from: <http://eprints.whiterose.ac.uk/780/>
19. Quinn AM, Drinkwater BW, Dwyer-Joyce RS. The measurement of contact pressure in machine elements using ultrasound. Available from: www.elsevier.com/locate/ultras
20. Zhou L, Brunskill H, Lewis R. Experimental investigation on ball plate contact using ultrasonic reflectometry: From static to dynamic. *Ultrasonics*. 2022 Aug 1;124. doi: 10.1016/j.ultras.2022.106733
21. Zhou L, Brunskill H, Pletz M, Daves W, Scheriau S, Lewis R. Real-Time Measurement of Dynamic Wheel-Rail Contacts Using Ultrasonic Reflectometry. *J Tribol*. 2019 Jun 1;141(6). doi: 10.1115/1.4043281
22. Dlair Obaid Ramadan. Ultrasonic Reflection Technique for Measuring Contact Conditions at the Tool Chip Interface. 2016.
23. Pau M, Aymerich F, Ginesu F. Ultrasonic measurements of nominal contact area and contact pressure in a wheel - rail system. *Proc Instn Mech Engrs V*. 2000;214(F).
24. Stratmann I, Goersch J, Schindler C. Overview of methods to identify the static normal wheel-rail contact. *Proc Inst Mech Eng F J Rail Rapid Transit*. 2022 Aug 1;236(7):783–92. doi: 10.1177/09544097211042487
25. Lekue J, Dörner F, Schindler C. On the Source of the Systematic Error of the Pressure Measurement Film Applied to Wheel-Rail Normal Contact Measurements. *J Tribol*. 2018 Mar 1;140(2). doi: 10.1115/1.4037358
26. Lekue J, Dörner F, Schindler C. Multiscale Finite Element Modeling of Wheel–Rail Rough Normal Contact Measurements Using Pressure Measurement Film. *Tribology Transactions*. 2018 Sep 3;61(5):972–8. doi: 10.1080/10402004.2018.1460433
27. Radmehr A, Ahangarnejad AH, Pan Y, Hosseini S, Tajaddini A, Ahmadian M. Wheel-Rail Contact Patch Geometry Measurement and Shape Analysis under Various Loading Conditions. In: *Joint Rail Conference*. St. Louis; 2020.
28. Vollebregt E. Detailed wheel/rail geometry processing with the conformal contact approach. *Multibody Syst Dyn*. 2021 Jun 1;52(2):135–67. doi: 10.1007/s11044-020-09762-w
29. Vollebregt E, Segal G. Solving conformal wheel-rail rolling contact problems. In: *Vehicle System Dynamics*. Taylor and Francis Ltd.; 2014. p. 455–68. doi: 10.1080/00423114.2014.906634
30. Radmehr A, Tajaddini A, Marquis B, Ahmadian M. Virginia Tech-Federal Railroad Administration Roller Rig Measurement Capabilities and Baseline Measurements. In: *Joint Rail Conference*. Snowbird, UT, USA; 2019.
31. Tactile Pressure Indicating Sensor Film Catalog. Fujifilm Prescale. 2015. Available from: www.Sensorprod.com

Table Titles and Figure Captions

TABLE 1 Test Condition.

TABLE 2 Type, Pressure Range, and Thickness of Pressure-sensitive Films.

FIGURE 1 Virginia Tech – Federal Railroad Administration (VT-FRA) Roller Rig.

FIGURE 2 Fujifilm Prescale Sheet Working Method (31).

FIGURE 3 (a) Front View of Contact Patch estimation method on the VT-FRA Roller Rig, (b) Before Experiment, (c) After Experiment.

FIGURE 4 Contact patch measurements for Low AoA using pressure-sensitive films.

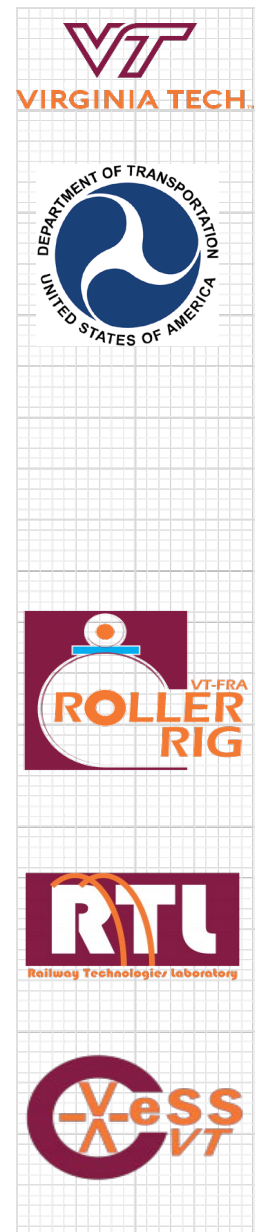
FIGURE 5 Contact patch measurements for Medium AoA using pressure-sensitive films.

FIGURE 6 Contact patch measurements for High AoA using pressure-sensitive films.

FIGURE 7 Contact Area Analysis for (a) L/V: 0.25; (b) L/V: 0.50; (c) L/V: 0.75; (d) L/V: 1.00.

FIGURE 8 Area Analysis for flange and tread contact point at higher AoA.

FIGURE 9 Initial CONTACT simulations to replicate static measurements.



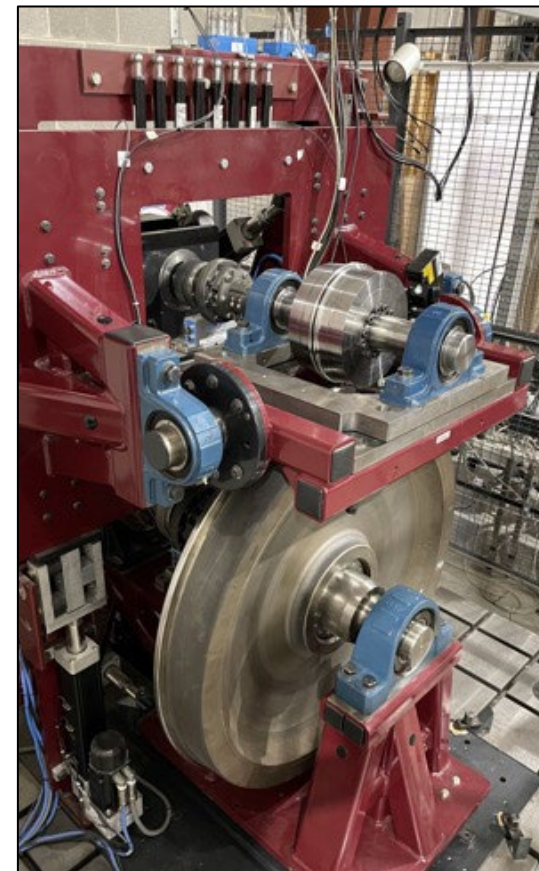
Estimation of Wheel-Rail Contact Patch Geometry Under Curving Conditions

Dr. Mehdi Ahmadian
J. Bernard Jones Chair and Director

Dr. Ahmad Radmehr
Research Scientist

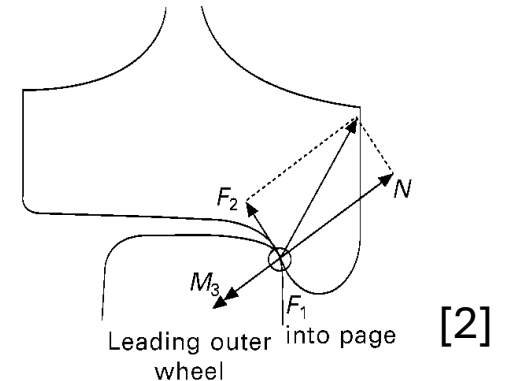
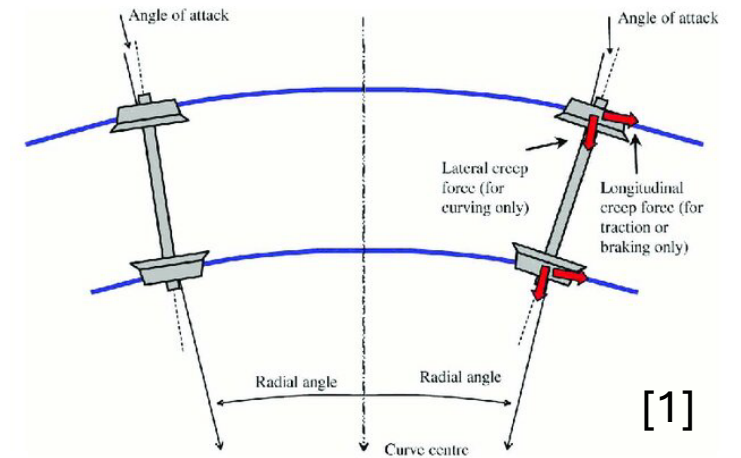
Nikhil Kumar
Graduate Research Assistant

Giovanni Mantovani
Graduate Research Assistant



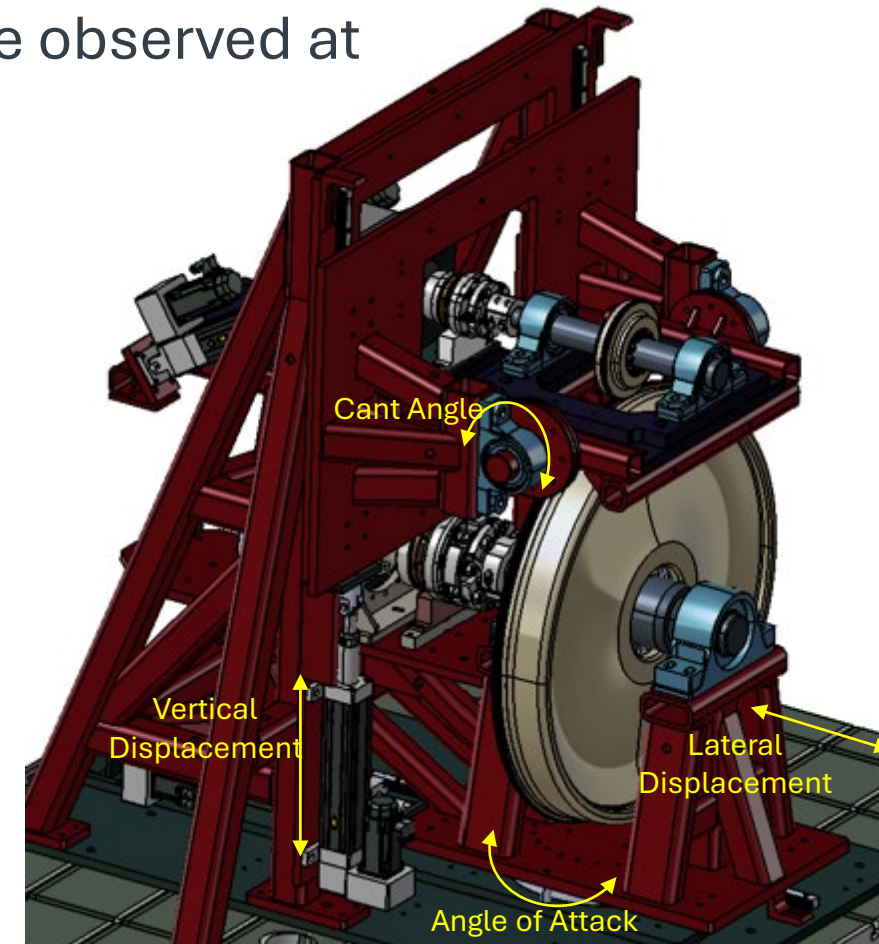
How does a curve affect the contact conditions of the WRI

- When a train negotiates a curve, the flange of the leading outer wheel contacts the rail
 - This creates a condition of two-point contact
- The railroad relies on various analytical and numerical-based contact models to help maintain the safety and standards of the industry
- Past modeling studies strongly suggest the need for experimental data for two-point contact conditions
- There is a big challenge in experimentally verifying the outcome of the several algorithms and software
 - To bridge this gap, the VT-FRA Roller Rig is deployed to access the contact patch under high angles of attack



VT-FRA Roller Rig Design Overview

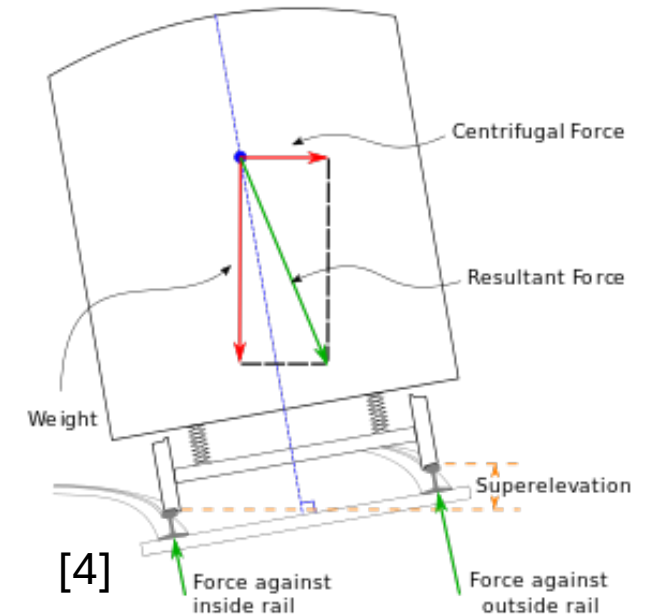
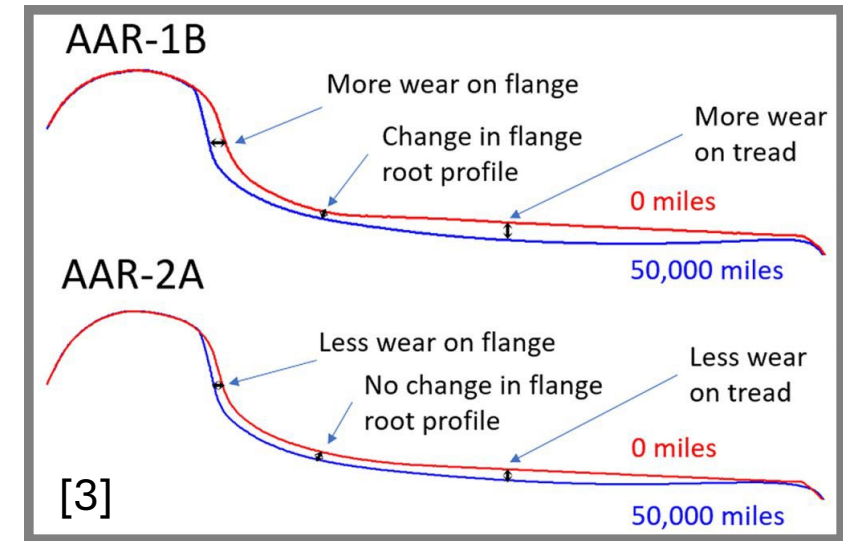
- A State-of-the-art, 1/4th scale roller rig that mimics the wheel and rail contact interface with a high level of precision
 - It helps better understand the complex dynamics that are observed at the WRI
- | | |
|--|---|
| ■ Two rotating bodies <ul style="list-style-type: none">○ Representing wheel and rail | ■ Two load measurement platforms <ul style="list-style-type: none">○ Each including four loadcells |
| ■ Six linear actuators, controlling <ul style="list-style-type: none">○ Wheel Load○ Angle of Attack (AoA) (Yaw)○ Lateral Displacement○ Cant Angle | ■ Precise laser profiler <ul style="list-style-type: none">○ Measuring surface changes and wear |
| ■ Two AC motors, independently driving <ul style="list-style-type: none">○ Wheel and Roller | ■ High-precision fluid dispensing system <ul style="list-style-type: none">○ For precise application of 3rd body layers |



Background and Testing Method

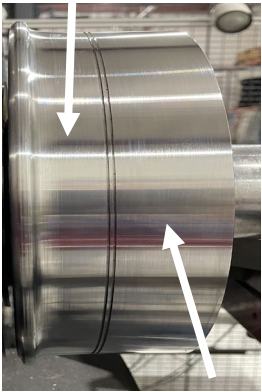
- AAR-2A wheel profile was introduced into the revenue service to reduce the impact of large lateral forces on the tread and flange
- The design of an AAR-2A profile was based on worn wheel profiles
 - To reduce wear by eliminating the harsh two-point contact
- Static measurements will help understand the possible contact locations, thus contact geometry and pressure distribution at high angles of attack

Testing Parameters	
Wheel Profile	AAR-2A
Roller Profile	RE-136
Wheel Load	5000 N
L/V Ratios	0.25,0.50,0.75,1.00
Rail Cant	1.5° and 3°
AoA	0°,1°,2°,3°,4°,5°



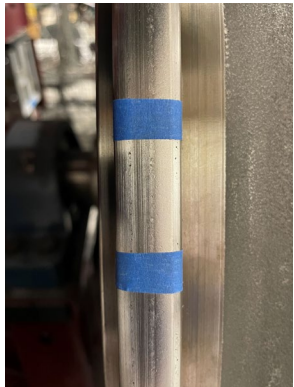
AAR-2A wheel profile and Pressure Sensitive Films are used to statically evaluate the contact patch

AAR-2A



Cylindrical

The test was performed on the redressed AAR-2A wheel surface



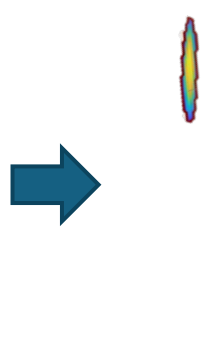
The contact patch was measured at the same location for all cant and AoA angles



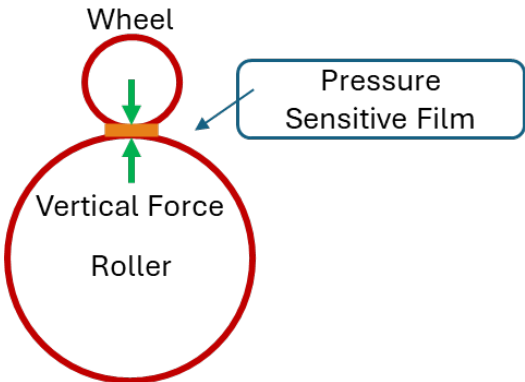
The cant angle was set for each measurement



Contact patch image



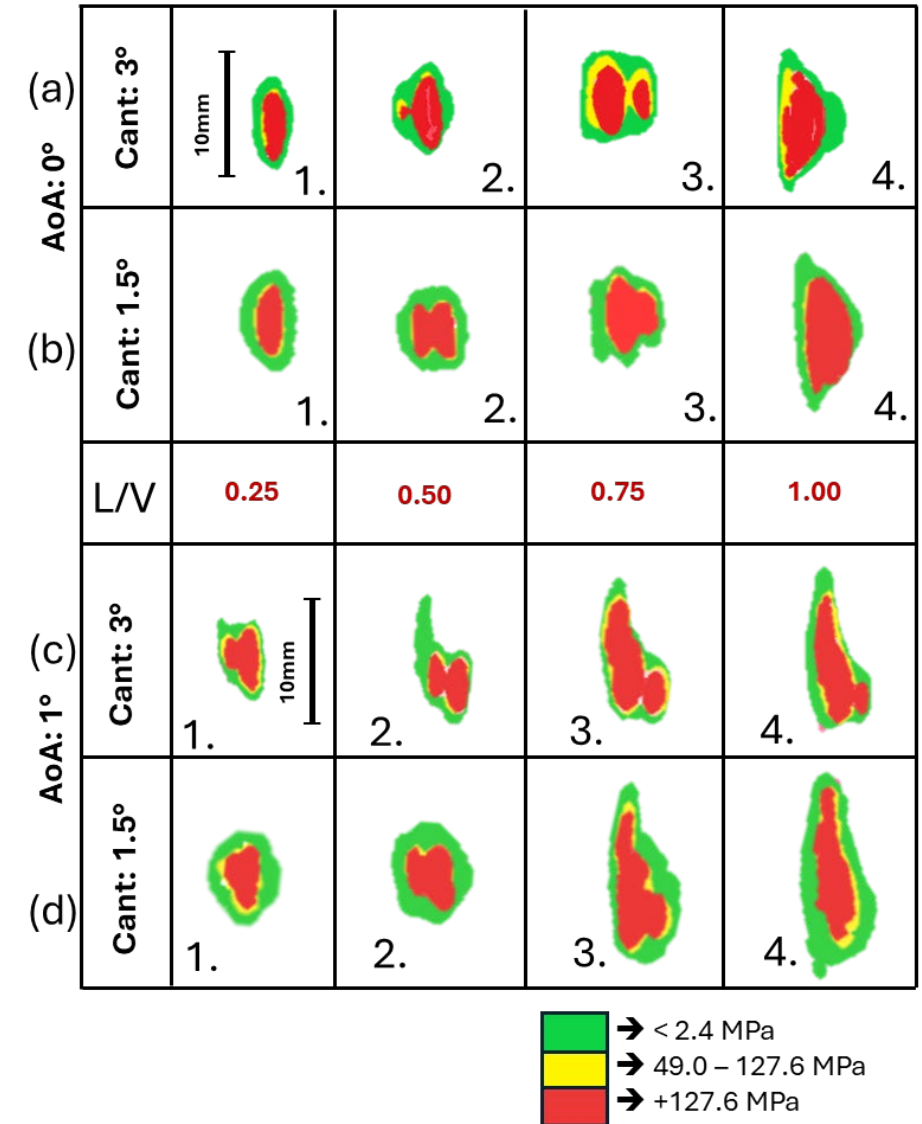
Contact simulation image



Film Type	Pressure Range	Thickness
Super-Low	0.5 – 2.4 MPa	Two-Sheet: 90 μ m each
High	49.0 – 127.6 MPa	Mono-sheet: 110 μ m
Super-High	127.6 – 297.9 MPa	Mono-sheet: 110 μ m

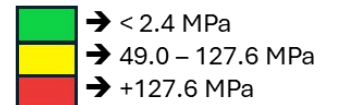
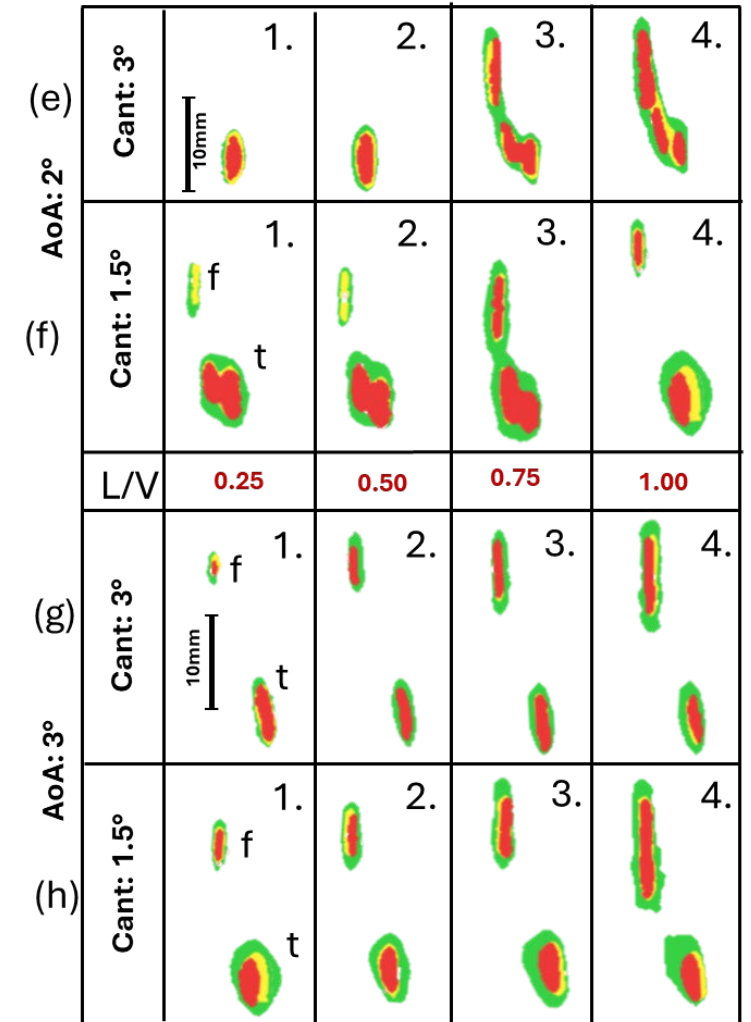
Conformal Contact: Impact of Lateral forces and low AoA

- Yaw angle is a result of the degree of curve
 - A higher degree curve results in a higher yaw, and a lower degree curve results in a lower yaw angle
- The Lateral and Vertical force ratio (L/V) is an important criterion for flanging conditions
 - Also called Derailment Coefficient
 - Lateral displacement of the outer wheel towards the rail causes a contact at the flange
- Smaller AoA results in a conformal contact at the wheel's throat
 - At lower L/V: The contact patches are more symmetrical and rounder (elliptical)
 - At higher L/V: The contact patches are more elongated and asymmetrical



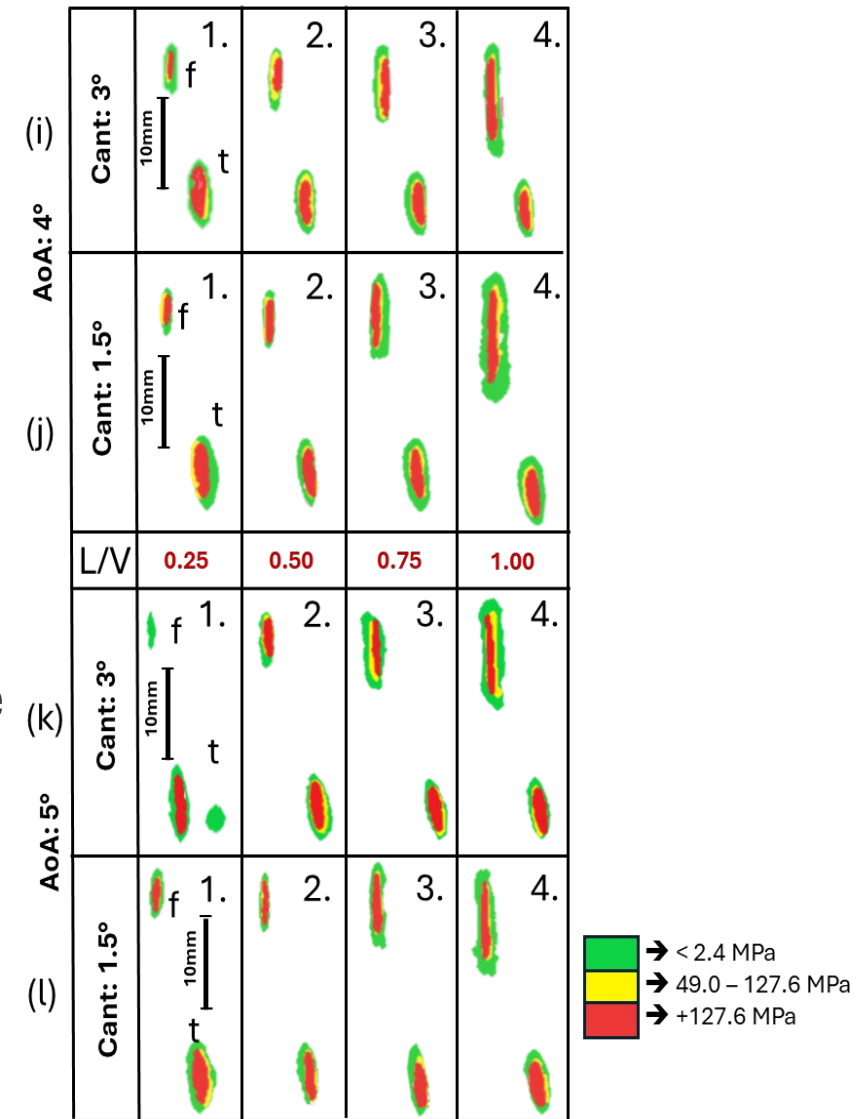
As AoA increases, the contact point shifts from one point to two points

- At moderate AoA, an onset of two-point contact is observed
- Higher cant angles (3°) resulted in a single point at lower lateral forces
 - Conforming to the throat at higher lateral forces
- With a change in rail cant, two-point contact occurs at lower lateral forces, but the higher-pressure concentrations remain on the tread
- Clear distinction of two-point contact is observed at AoA 3°
 - The distance between the two points is estimated to be ≈ 10 mm
 - The aspect ratio of the flange contact point increases with increasing flange contact
- Visually, rail cant of 1.5° appears to have a larger contact area when compared with 3°



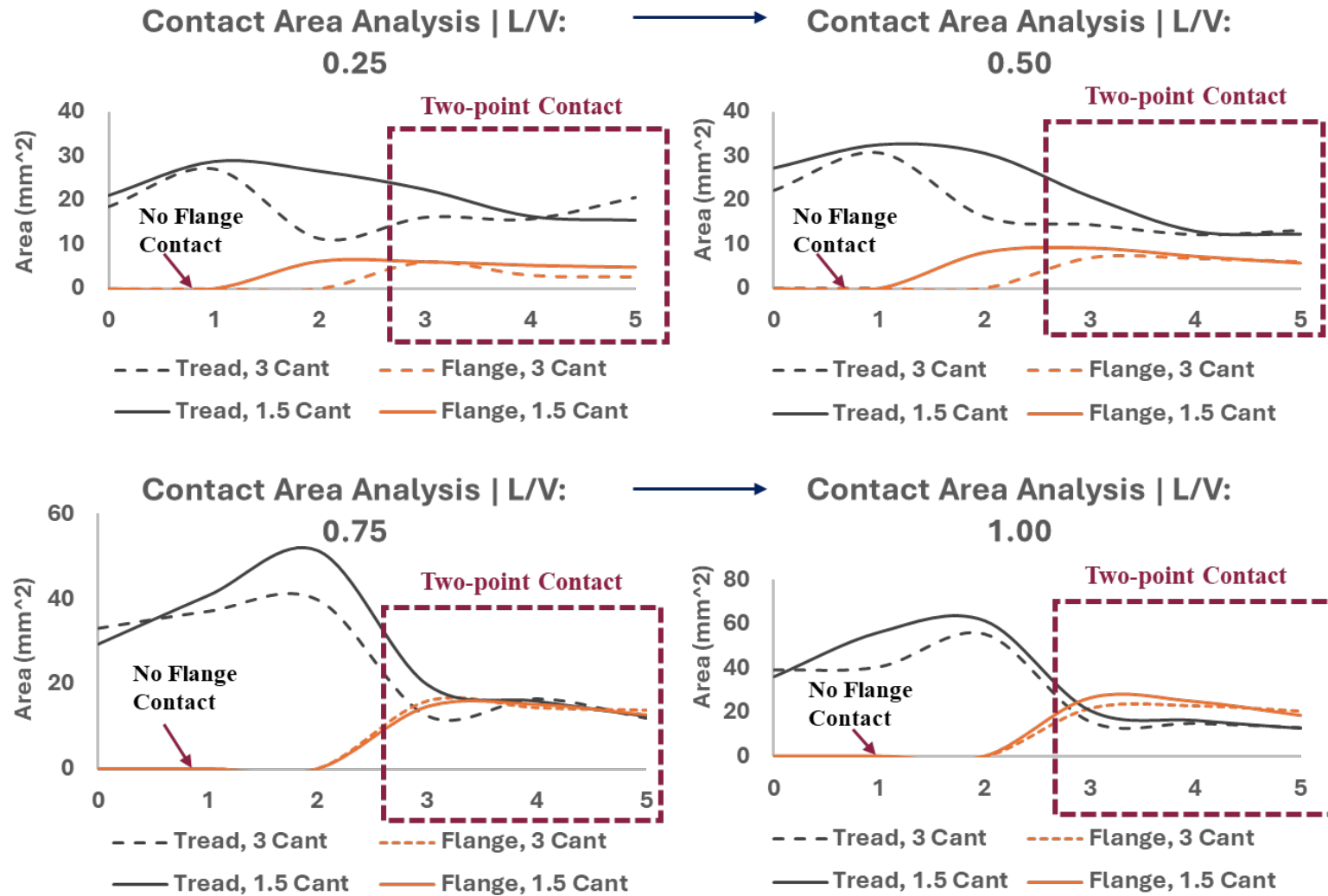
L/V ratios influence the area and pressure distribution at the flange

- The AAR-2A wheel profile was designed with the intention of having lesser wear on the flange and the tread owing to a reduction in two-point contact conditions
 - As the L/V ratio increases
 - The aspect ratio of the flange contact point increases
 - The distance between the two points decreases
 - The distance between the two points is greater than 10 mm
- Pressure-sensitive films provide an experimental understanding of flanging contact conditions, such as the pressure distribution and geometry
- CONTACT simulations, in comparison to the static measurements, will assess the model's ability to predict the contact points
 - To relate the results to steel-to-steel contact



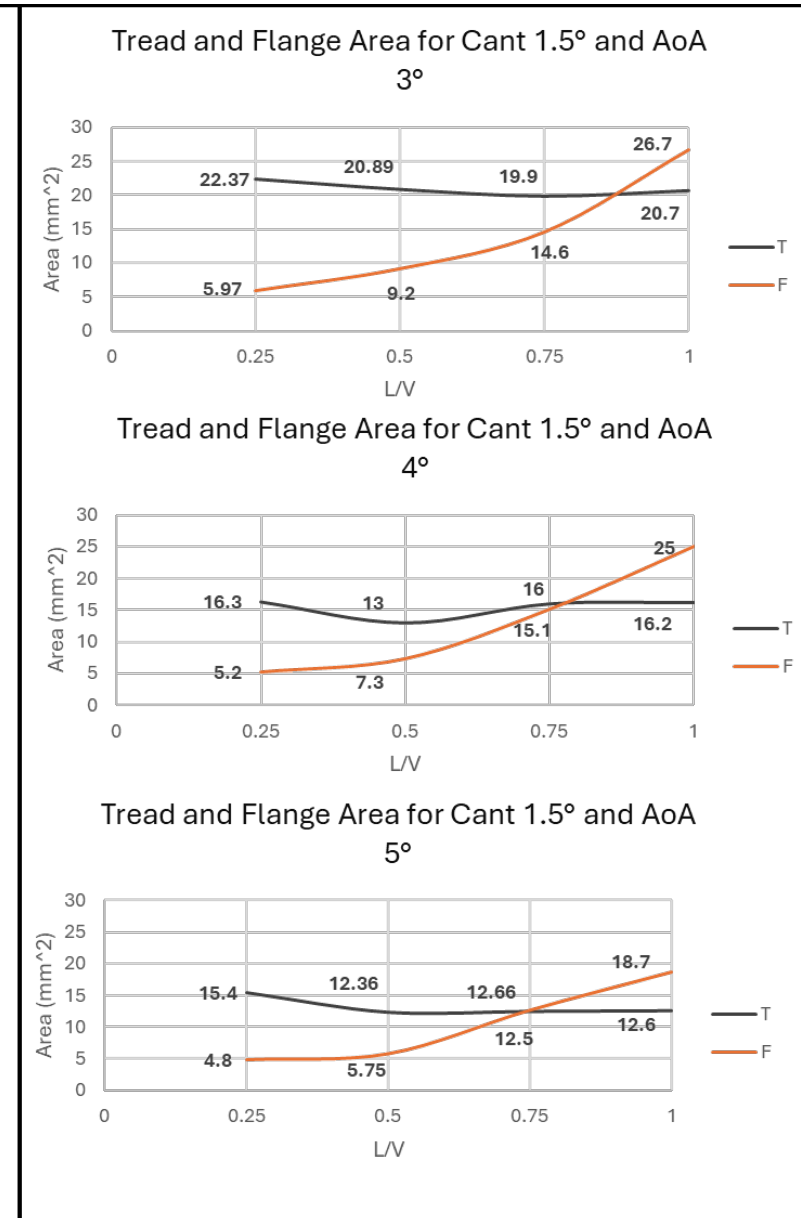
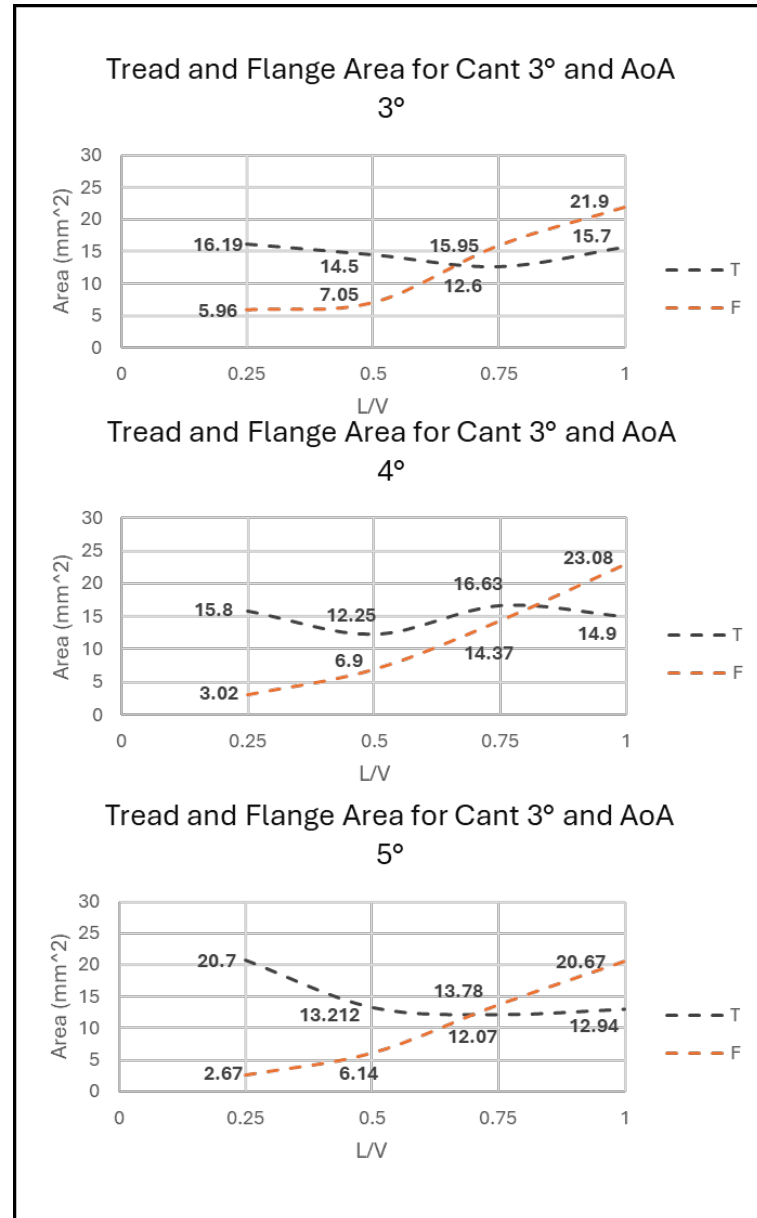
Contact area comparison for each tread contact area at different L/V ratio

- At lower AoA, the contact area is restricted to the tread of the wheel
 - A similar pattern observed under all L/V conditions
- The plots clearly identify regions of two-point contact, which occur primarily at higher AoA values (3° to 5°) across all L/V ratios
- While tread contact area dominates at lower AoA, flange contact becomes more prominent at higher AoA. This shift is crucial for maintenance planning, as it indicates potential wear zones



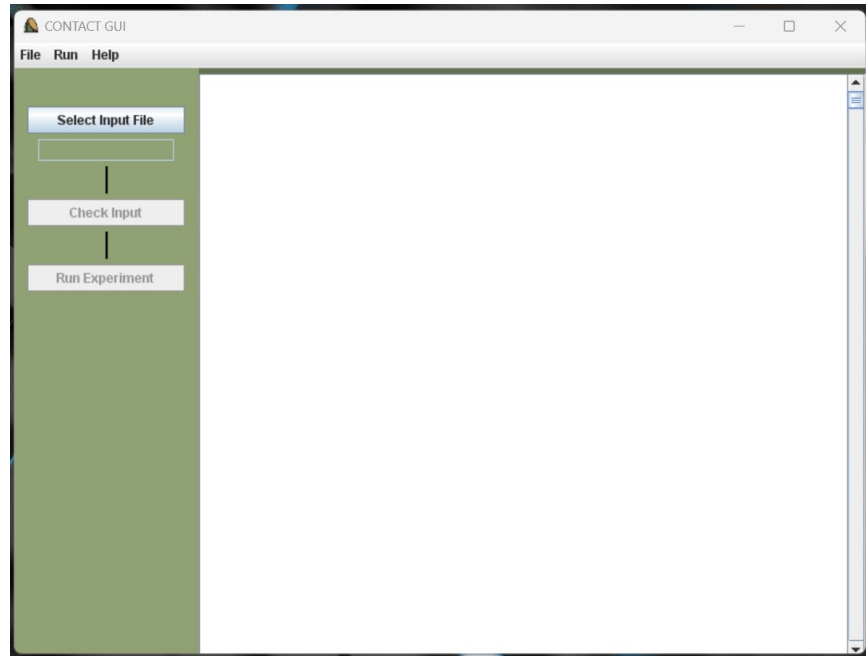
Flange and tread area comparison at different L/V ratios for higher AoA

- At lower lateral to vertical force ratios (L/V), the tread contact area is generally higher than the flange contact area
- A shift in the contact stress from the tread to the flange with increasing lateral forces is noticed
 - Tread contact area remains approximately constant
- The flange contact area increases non-linearly, suggesting a more complex interplay of contact mechanics

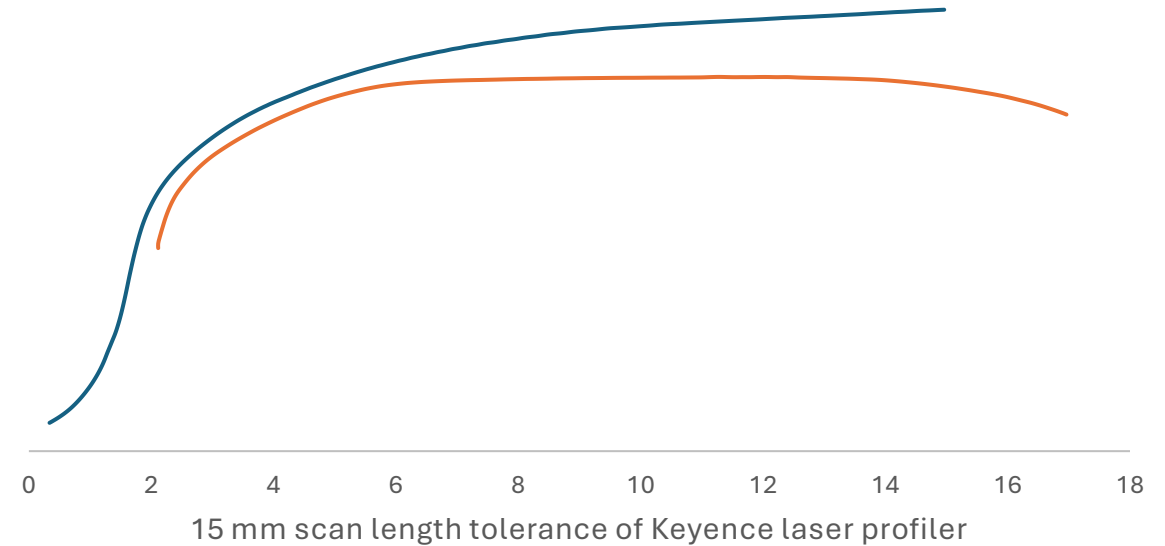


CONTACT software is used to simulate the intricate dynamics of WRI

CONTACT GUI



[5]



- The Keyence laser scanner was used to scan the roller and wheel topography
 - The blue line shows the AAR2A wheel profile used
 - The orange line depicts the roller profile

A “Flexibility” parameter is used to evaluate the effect of a compliant third-body film at the wheel/roller (rail) contact

- For the simulations, a new line of code was added to incorporate the effect of a flexible film-like 3BL

$$L_z = h^{(3)} \frac{(1 + \nu^{(3)})(1 - 2\nu^{(3)})}{(1 - \nu^{(3)})E^{(3)}}.$$

- Polyester Film Mechanical Properties^[6]:
 - E = 92.8 N/mm²
 - ν = 0.48 (-)
 - h = 0.2 mm
- Using the equation above:
 - FLX = 0.00024 (mm³/N)
- Simulations were performed at both 3° (+/- 0.1) and 1.5° (+/- 0.1)

```
% Contact Patch Investigation for Two Point Flange Contact
% This Document is the input file for:
% Fz = 5 kN, Cant = 3°, AoA = 5°
% Natural 3BL = Not present
% FLX = Presence of elastic film material

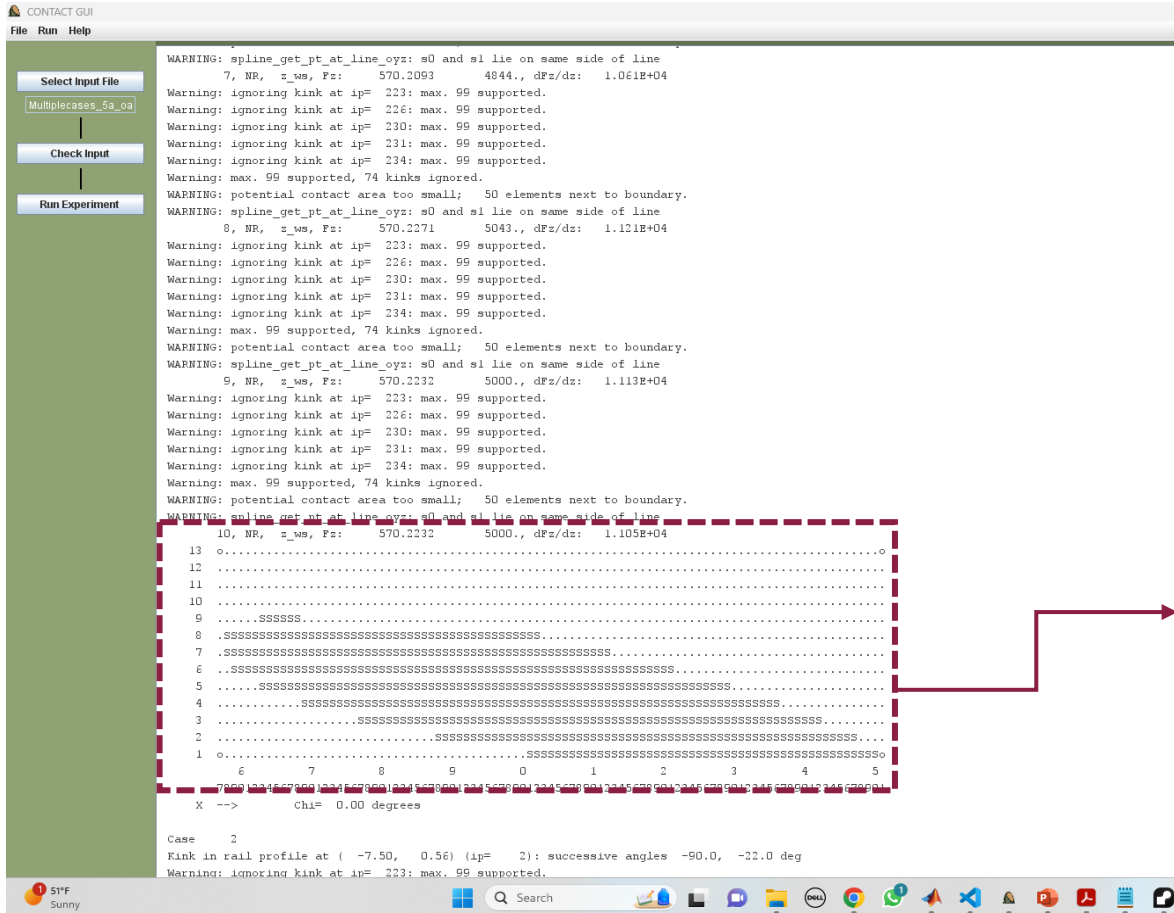
1 MODULE
0210100 C-P-B-T-N-F-S CONFIG, PVTIME, BOUND, TANG, NORM, FORCE, STRESS
0022433 V-L-D-C-M-Z-E VARFRC, FRCLAW, DISCNS, INFLCF, MATER, ZTRACK, EWHEEL
0001521 H-G-I-A-O-W-R HEAT, GAUSEI, IESTIM, MATFIL, OUTPUT, FLOW, RETURN
199 100 30 1 1e-5 MAXGS, MAXIN, MAXNR, MAXOUT, EPS
0.550 0.550 FSTAT, FKIN
0.280 0.280 82000. 82000. POISS 1,2, GG 1,2
0.0002453014 FLX
500.0 .0020 480.0 12000. GG3, LAYTHK, TAU_C0, K_TAU
0.100 0.100 1.000 180d 8.0 4.0 DX, DS, DQREL, A_SEP, D_SEP, D_COMB
0.000 -1.0 571.5 0 0 GAUGWD, GAUGHT, NOMRADR, RAILYO, RAILZO

'Arema136_v2_11162023.prr' 0 1. 0. RFNAME, MIRRORV, SCALE, SMOOTH
0.0 0.0 0.000 0.0 0.0 0.000 DY, DZ, ROLL, VY, VZ, VROLL
-38.894 0.0 127.0 % 10"/2 FBDIST, FBPOS, NOMRADW
'AAR2A_inv_11162023.prw' 0 1. 0. WFNAME, MIRRORV, SCALE, SMOOTH
0.0 -17.991 5000. 1.6d 5d 0.0 S, Y, FZ, ROLL, YAW, PITCH
0.0 0.0 0.0 0.0 0.0 0.0 RPITCH, VY, VZ, VROLL, VYAW, VPITCH

0 MODULE
```

Sample Input File

A sample of CONTACT Output



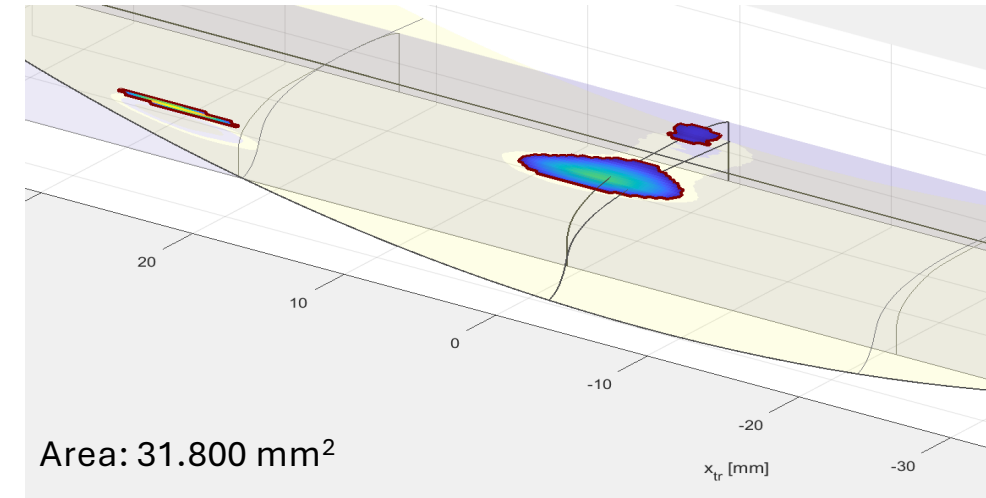
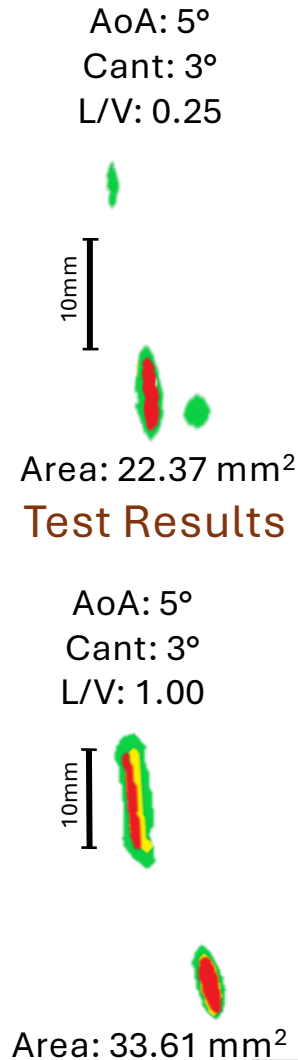
```
1 clear all
2 clc
3 close all
4 addpath('C:\Program Files\Vtech CMCC\contact_v23.1\matlab');
5 addpath('C:\Program Files\Vtech CMCC\contact_v23.1\matlab_intf');
6 l1 = loadcase('Multiplecases_5a_oa',3);
7 % l2 = loadcase('ContactPatch-AAR2A_0AoA',2);
8 prr = read_profile('Arema136_v2_11162023.prr');
9 prw = read_profile('AAR2A_inv_11162023.prw');
10 opt = plot3d; opt.rw_surfc = 'both';
11 % opt_w = plot3d; opt_w.rw_surfc = 'prw';
12 figure (1)
13 plot3d( l1, opt, prr, prw );
14 % figure (2)
15 % plot3d( l2, opt, prr, prw );
16 % figure (1)
17 % plot3d(l13, opt, prr);
18 % figure (2)
19 % plot3d(l13, opt_w, prw);
20 area = sum(sum(l1.pn>0))*l1.dx*l1.dy
21 % area2 = sum(sum(l2.pn>0))*l2.dx*l2.dy
22 % figure(2)
23 % contourf(l13.y,l13.x,l13.py')
24
25 % Fx = sum(sum(l13.px*l13.dx*l13.dy))
26 % Fy = sum(sum(l13.py*l13.dx*l13.dy))
27 % Fn = sum(sum(l13.pn*l13.dx*l13.dy))
```

- The contact patch
- The data is exported to Matlab to create a better visualization
- Plot3d is used to plot the contact patch in 3d and visualize both the wheel and rail profiles

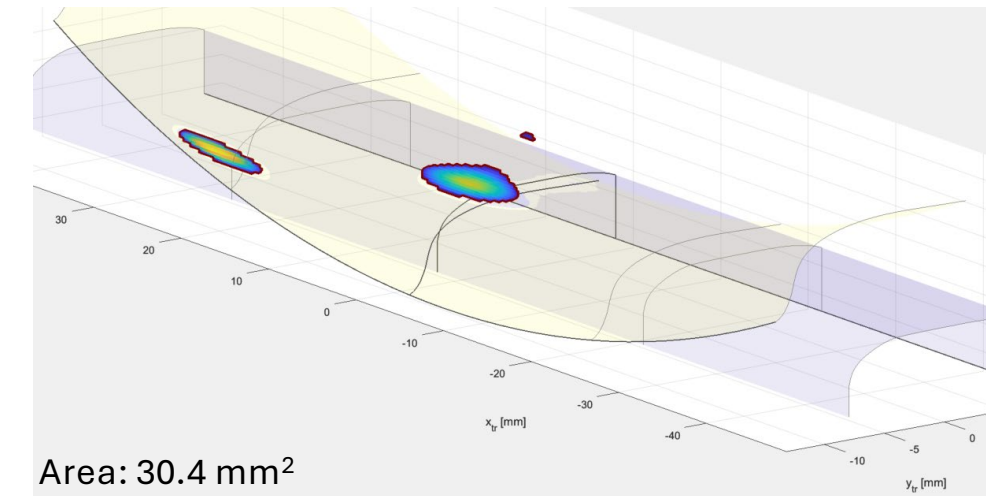


Initial simulations show consensus to the static measurements with the pressure-sensitive film

- CONTACT is able to predict the contact patches for the wheel and roller profiles of the Roller Rig
- The anomalous contact point due to the wheel profile in the test results is precisely captured through modeling
- Tuning of the model and steel-to-steel simulations are necessary for further validating the model

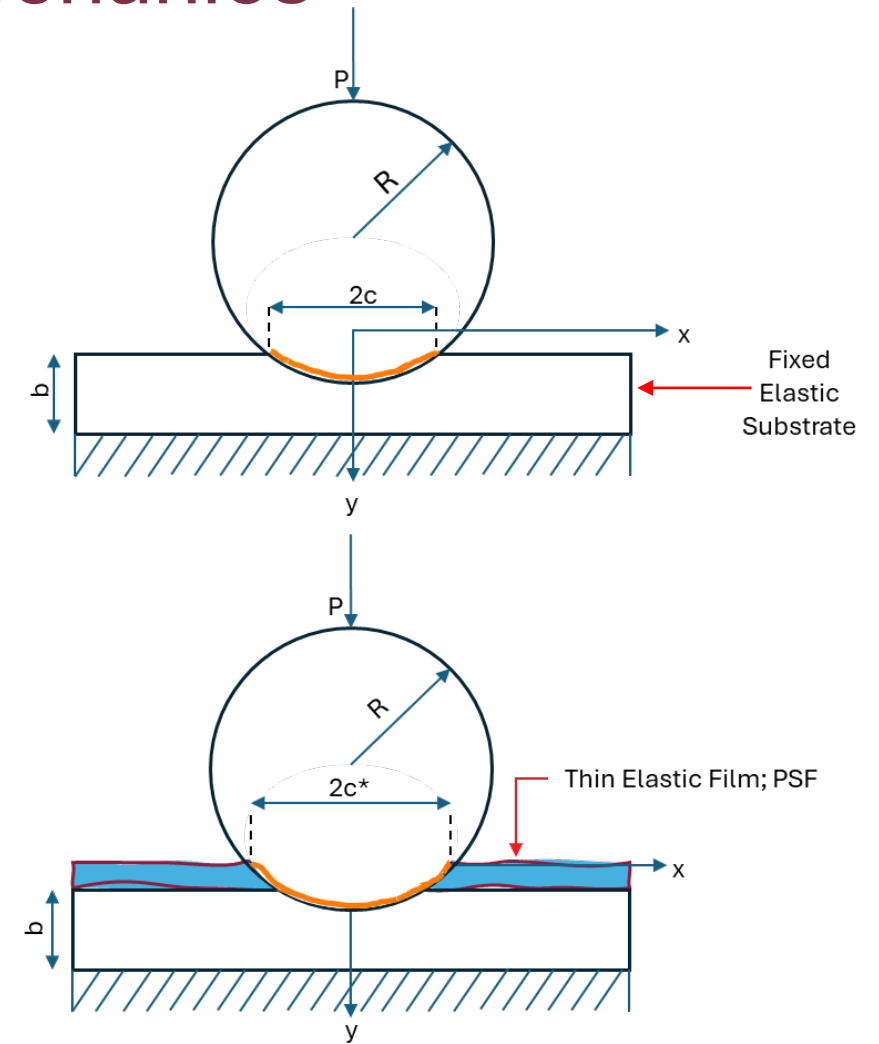


CONTACT Modelling



The reduction factor helps validate the contact model and theories related to wheel-rail contact mechanics

- A thin elastic film at contact increases the contact patch area (orange) and follows a typical Hertzian contact pressure distribution (also used in the simulation algorithm)
- The pressure distribution decreases while the contact area increases with the presence of an elastic film
- This difference in the area is estimated to be of a magnitude of **8 to 9 times** steel-to-steel contact
- Tuning of the model is necessary to get accurate results, which is in the future scope of this project



Conclusions

- The results highlight that the location and centroid of contact pressure are highly affected by the relative lateral position and angle of attack between the wheel and rail
- Beyond what had been reported in the past modeling studies
- These static measurements help us understand the geometry of the contact and provide a method to validate the simulation results
- The results of the study could be helpful in better assessing curving conditions that can cause derailments
- Comparisons with CONTACT show agreement in contact patterns
- Future Scope:
 - Tuning of the CONTACT model to have better agreement with lesser percent difference in the geometry
 - Steel-to-steel contact evaluation and scaling for field condition
 - Possible Dynamic experimentation and simulation for accessing the L/V ratios under the different curving conditions



References

1. Wei, X., Zhu, M., & Jia, L. (2016). A semi-active control suspension system for railway vehicles with magnetorheological fluid dampers. *Vehicle System Dynamics*, 54(7), 982–1003.
<https://doi.org/10.1080/00423114.2016.1177189>
2. Thompson DJ, Jones CJC. Noise and vibration from the wheel-rail interface. In: Lewis R, Olofsson O (eds). *Wheel/rail interface handbook*. Oxford, UK: Woodhead Publishing Ltd, 2009; 477–509.
3. <https://www.railwayage.com/news/next-generation-wheel-profile-aar-2a/>
4. Garritsen, K. (2019). Variable static speed profiles within ERTMS on the Dutch rail network. Investigation of the possibilities and implementation of variable static speed profiles based on external characteristics.
5. Vollebregt E. User guide for CONTACT, Rolling and sliding contact with friction. VORtech; 2017. Report No. TR09-03, version 17.1; see www.kalkersoftware.org.
6. <https://www.matweb.com/search/DataSheet.aspx?MatGUID=40559706b4fd4aa0a43f5739799728f5&ckck=1>



Thank You.

We want to acknowledge the Federal Railroad Administration, DOT, for funding this project.



Backup Slides



VIRGINIA TECH.



Plot3D function in MATLAB allows us to visualize the contact patch/point for the actual wheel and roller profiles

- The effect of a film 3BL is added in CONTACT simulations
- The image on the left is from the pressure-sensitive films
- The image on the right is from simulations
- Interestingly, there is general agreement between the contact profile from tests and simulations
 - A more in-depth analysis is needed to better understand the differences in total area from testing and modeling

